MINOR PROJECT – 4

PROJECT INTRODUCTION

Project Title : Employee Attrition Prediction Using Random Forest

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Dataset Name: IBM HR Analytics Employee Attrition & Performance Dataset

Dataset Link : https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Short Description of the Datase: The dataset contains 1,470 employee records with 35 attributes, including age, job role, salary, job satisfaction, overtime, and years at the company. The target variable, Attrition, indicates whether an employee has left the organization (Yes) or stayed (No).

Objective of the Analysis: The objective of this project is to develop a Random Forest Classification model to predict employee attrition. The analysis involves data exploration, handling missing values and duplicates, removing irrelevant columns, encoding categorical variables, splitting the data into training and testing sets, and training the model. Performance is evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, Confusion Matrix, and ROC Curve. Feature importance is also analyzed to identify the key factors contributing to predictions.

Model Used: Random Forest Classifier

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, Confusion Matrix, and ROC Curve.

Conclusion

This project demonstrates the use of Random Forest for predicting employee attrition. It helps identify patterns in employee data and understand the factors contributing to attrition predictions, supporting data-driven employee retention decisions.